3月21日

微积分，共3题。

请务必休息一天

祝顺利！

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1. $\lim\limits_{x\to+\infty}(x^{\frac{2}{x}}-1)^{\frac{1}{\ln x}}=$

考虑 $\lim\limits_{x\to+\infty}\dfrac{2\ln(x^{\frac{2}{x}}-1)}{\ln x} \overset{洛}{=} \lim\limits_{x\to+\infty}\dfrac{2\frac{2-2\ln x}{x^2}e^{\frac{2}{x}\ln x}}{\frac{1}{x}(x^{\frac{2}{x}}-1)} = \lim\limits_{x\to+\infty}\dfrac{2(2-2\ln x)}{x(x^{\frac{2}{x}}-1)}$

$= \lim\limits_{x\to\infty}\dfrac{2(2-2\ln x)}{x(\frac{2}{x}\ln x)} = \lim\limits_{x\to+\infty}\dfrac{4-4\ln x}{2\ln x} = -2.$

故 $I = e^{-2}$

2.设 n 为非负整数，则 $\int_0^1 x^2\ln^n x\mathrm{d}x=$

$I_n = \frac{1}{3}\int_0^1 \ln^n x\, dx^3 = \frac{1}{3}x^3\ln^n x\Big|_0^1 - \frac{1}{3}\int_0^1 x^3 n\ln^{n-1}x\,\frac{1}{x}\,dx.$

$= -\frac{n}{3}\int_0^1 x^2\ln^{n-1}x\,dx = -\frac{n}{3}I_{n-1} = \cdots = (-1)^n\frac{n!}{3^n}I_0$

$I_0 = \int_0^1 x^2\,dx = \frac{1}{3}$

故 $I_n = (-1)^n\frac{n!}{3^{n+1}}$

3.设 $F(x)>0$ 为 $\mathbb{R}$ 上的连续可导函数，$F(0)=\sqrt{\pi}$，且 $F(x)F'(x)=\frac{\cos x}{2\sin^2x+\cos^2x}$，求 $F(x)$

记 $F'(x) = \frac{dF}{dx}$，$F(0)=\sqrt{\pi}$

$F\cdot\frac{dF}{dx} = \frac{\cos x}{2\sin^2x+\cos^2x}$

$\int F\,dF = \int\frac{d\sin x}{1+\sin^2x}$

$\frac{1}{2}F^2 = \arctan\sin x + C$，又 $C = \frac{\pi}{2}$

故 $F = \sqrt{\arctan\sin x + \pi}$